gpeyt1-46 · Driven pile in clay (α)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 16 in. square precast concrete pile is driven 50 ft into a two-layer clay deposit: 0–20 ft soft clay su = 600 psf (α = 0.95); 20–60 ft stiff clay su = 1400 psf (α = 0.62). The water table is at 5 ft. Clay γsat = 115 pcf. Use the α-method for side resistance and qp = 9su at the toe. Design load 90 kips; ASD FS = 2.5.

Most nearly, the ultimate axial compression capacity Ru is:

A. 310 kips
B. 200 kips
C. 174 kips
D. 222 kips

Given equations / data

> Rs = Σ α su (perimeter)(ΔL); Rt = 9 su At; Ru = Rs + Rt

Worked solution

1. Perimeter = 4×1.333 = 5.333 ft; At = 1.778 ft²
2. Layer 1: 0.95×600×5.333×20 = 60,800 lb
3. Layer 2: 0.62×1400×5.333×30 = 138,880 lb
4. Rt = 9×1400×1.778 = 22,400 lb
5. Ru = 222,080 lb ≈ 222 kips (Rall = 89 kips at FS 2.5 > 90 ✓)

Final answer

222 kips

Why the other values are wrong

- **310 kips** — omitted α (α = 1)
- **200 kips** — left out toe resistance
- **174 kips** — treated the pile as a 16 in. round pile

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 α-method (Eq. 7-31) and qp = 9su (Eq. 7-16)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=hfbcwIJfAjw, https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=ZcjQbEB66IQ (no creator values copied).

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gpeyt1-47 · Driven pile in sand (β)

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Problem

A 14 in. diameter closed-end steel pipe pile is driven 45 ft into medium-dense sand. Water table at 10 ft; γ = 115 pcf above, γsat = 125 pcf below. Use the effective-stress β-method with β = 0.38 along the full length and toe coefficient Nt = 40 (from the Fellenius ranges for this sand). Ignore any limiting values. SPT N60 ≈ 22; φ′ ≈ 34°.

Most nearly, the ultimate capacity Ru is:

A. 407 kips
B. 260 kips
C. 117 kips
D. 352 kips

Given equations / data

> Rs = β ∫σ′v dA (use average σ′v of each segment × side area); Rt = Nt σ′v,toe At; γw = 62.4 pcf

Worked solution

1. σ′v at 10 ft = 1,150 psf; at toe = 1,150 + 35(125−62.4) = 3,341 psf
2. Perimeter = π(1.167) = 3.665 ft; At = 1.069 ft²
3. Σσ′v·L = ½(1,150)(10) + ½(1,150+3,341)(35) = 84,342 lb/ft
4. Rs = 0.38×3.665×that = 117,470 lb
5. Rt = 40×3,341×1.069 = 142,863 lb
6. Ru = 260 kips

Final answer

260 kips

Why the other values are wrong

- **407 kips** — used total stress (no pore-pressure subtraction)
- **117 kips** — left out toe resistance
- **352 kips** — used toe σ′v along the whole shaft

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7.2 effective-stress (β) method, Rs and Rt with Nt

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Jdcs8-i_OA4, https://www.youtube.com/watch?v=hfbcwIJfAjw (no creator values copied).

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gpeyt1-48 · Uplift capacity in clay

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A transmission tower leg is founded on an 18 in. diameter drilled concrete pier, 40 ft long, in uniform stiff clay (su = 1,100 psf, adhesion factor α = 0.7). The water table is deep (below the pier). Concrete γc = 150 pcf. Apply FS = 3.0 to side resistance only; pier self-weight is counted at full value (unfactored). Ignore any belling and end suction.

Most nearly, the allowable uplift capacity is:

A. 52 kips
B. 59 kips
C. 48 kips
D. 55 kips

Given equations / data

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Tall = α su π D L / FS + Wpier
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Worked solution

1. Qs = 0.7×1,100×π×1.50×40 = 145,142 lb
2. W = π/4×1.50²×40×150 = 10,603 lb
3. Tall = 145,142/3.0 + 10,603 = 58,983 lb ≈ 59 kips

Final answer

59 kips

Why the other values are wrong

- **52 kips** — applied FS to pier weight too
- **48 kips** — ignored pier weight
- **55 kips** — used buoyant concrete weight though WT is deep

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) α-method side resistance; uplift Rn = Rs + W (Eq. 7-41 group form)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=jnAwn45noZw (no creator values copied).

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gpeyt1-49 · Pile group block failure

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A 3 × 3 group of 12 in. square piles at 3 ft center-to-center spacing is driven 45 ft into soft-to-medium clay with weighted average su1 = 900 psf along the embedment and su2 = 1,600 psf below the toe (to 2B). Each single pile has nominal capacity 95 kips (α-method + toe). Pile cap is not in contact with the ground.

Most nearly, the nominal group capacity is:

A. 1,840 kips
B. 855 kips
C. 1,490 kips
D. 1,273 kips

Given equations / data

> Block: Rng = 2D(B + Z)su1 + B Z su2 Nc; Nc = 5(1 + D/5B)(1 + B/5Z) ≤ 9; B = Z = outside-to-outside group width; group capacity = lesser of block and Σ singles

Worked solution

1. B = Z = 2(3) + 1 = 7 ft; Nc = 5(1 + 45/35.0)(1.2) = 13.7 → cap at 9
2. Rblock = 2(45)(7.0+7.0)(900) + 7.0²(1,600)(9) = 1,839,600 lb
3. Σ singles = 9 × 95 = 855 kips
4. Group = lesser = 855 kips

Final answer

855 kips

Why the other values are wrong

- **1,840 kips** — took the larger value
- **1,490 kips** — used center-to-center (6 ft) group width for block
- **1,273 kips** — forgot the 2 on perimeter shear

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I Eq. 7-36 and 7-37 (block failure, Nc)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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gpeyt1-50 · Pile group settlement (equivalent footing)

GEO PE · Long Calculation Problems I — Topic 10: Deep Foundations

Problem

A pile group with plan dimensions 8 ft × 8 ft (outside-to-outside) carries Q = 900 kips. The friction piles are 45 ft long in a stiff clay crust that overlies a 20 ft normally consolidated clay layer from 45 ft to 65 ft (γsat = 118 pcf, e0 = 1.0, Cc = 0.36). Water table at the ground surface; stiff crust also γsat = 118 pcf. Place the equivalent footing at 2/3 of the embedded length and spread load at 2:1; evaluate one layer at mid-depth of the NC clay.

Most nearly, the consolidation settlement of the group is:

A. 4.5 in.
B. 12.1 in.
C. 1.3 in.
D. 2.2 in.

Given equations / data

> Equivalent footing at D′ = (2/3)D; Δσ = Q/[(B + z)(L + z)] with z measured from D′; S = Cc H/(1 + e0) log(σ′f/σ′0)

Worked solution

1. D′ = (2/3)(45) = 30 ft; mid-clay depth = 55 ft → z = 25 ft
2. Δσ = 900,000/(8+25)² = 826 psf
3. σ′0 = 55(118−62.4) = 3,058 psf
4. S = 0.36×20/2.0 × log(3,884/3,058) = 4.5 in.

Final answer

4.5 in.

Why the other values are wrong

- **12.1 in.** — placed the equivalent footing at the pile toe
- **1.3 in.** — spread the load from the ground surface
- **2.2 in.** — used total stress for σ′0

References (Jayce's library)

- FHWA-NHI-16-009 (GEC 12) Vol I §7 equivalent-footing method (2/3 D)
- NCEES PE Civil Reference Handbook v2.1 §3.5.1 2:1 method; §3.2.1

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=MI4gaUwboyE (no creator values copied).

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